

The Thom Isomorphism and Characteristic Classes

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1 Homology and cohomology

2 Smooth vector bundles

2.1 Definitions and basic properties

2.2 Oriented vector bundles

Let $\pi: E \rightarrow B$ be an oriented rank- n vector bundle. The Thom isomorphism relates the cohomology $H^k(B; R)$ to the relative cohomology of $H^{k+n}(E, E^\circ; R)$ where E° is the complement of the zero section of E and R is an (associative) unital ring. Note that for each fiber $F = \pi^{-1}(b)$, $b \in B$, with $F^\circ = F \cap E^\circ$ being the nonzero elements of F , the relative cohomology $H^*(F, F^\circ; R)$ is equal to

$$H^*(F, F^\circ; R) = \begin{cases} R & \text{if } * = n \\ 0 & \text{otherwise;} \end{cases}$$

one can see this by considering the long exact sequence of the pair (F, F°) . The orientation on the vector bundle then confers a canonical generator $u_{F, \mathbb{Z}}$ upon each fiberwise cohomology group $H^n(F, F^\circ; \mathbb{Z}) \cong \mathbb{Z}$. Then, since there is a unique homomorphism $\mathbb{Z} \rightarrow R$ for any unital ring R , we may define $u_{F, R}$ to be the image of $u_{F, \mathbb{Z}}$ under the map $H^n(F, F^\circ; \mathbb{Z}) \rightarrow H^n(F, F^\circ; R)$; thus the orientation of the vector bundle gives a canonical generator for the fiberwise cohomologies regardless of the ring.

3 Thom isomorphism

We follow Milnor and Stasheff's account of the Thom isomorphism theorem.

Definition 3.1 (Thom class) — A *Thom class* $u = u_R \in H^n(E, E^\circ; R)$ which restricts to $u_F = u_{F, R}$ for every fiber $F = \pi^{-1}(b)$ above a point $b \in B$.

Theorem 3.2 (Thom isomorphism). Each R -oriented vector bundle $\pi: E \rightarrow B$ possesses a unique Thom class. Furthermore, there is an isomorphism

$$\begin{aligned} \Phi: H^k(E; R) &\rightarrow H^{k+n}(E, E^\circ; R) \\ y &\mapsto y \smile u. \end{aligned}$$

Note that E deformation retracts onto the zero section, which is homeomorphic to B , by simply linearly contracting each fiber. Thus the Thom isomorphism gives an isomorphism $\Phi \circ \pi^*: H^k(B; R) \rightarrow H^k(E, E^\circ; R)$.

Some geometric intuition for the Thom class: Consider the space of differential n -forms on E whose restriction to each fiber is compactly supported; these are called *differential forms with compact vertical supports*, which form cohomology groups $H_{cv}^*(E)$. A *Thom form* ϕ is an n -form on E whose restriction to each fiber integrates to 1 (thus represents the Thom class of E when $R = \mathbb{R}$). The Thom isomorphism theorem says that there is an isomorphism $H_{dR}^k(B) \rightarrow H_{cv}^{k+n}(E)$ via wedging with ϕ ; the inverse of this map is given by integrating a compactly vertically supported $(k+n)$ -form on each fiber to get a k -form on B .

The proof of Theorem 3.2 will proceed in roughly three steps: first, we prove it for trivial vector bundles; then, we prove it when B can be covered by finitely many open sets over which the given vector bundle is trivial; and finally, we prove it in the general case by taking a limit over compact sets.

The case of a trivial bundle.

The case of a good finite cover of B . We will show that if that $B = B_1 \cup B_2$ is a cover of B by two open sets such that Theorem 3.2 holds for the restricted vector bundles $\pi|_{\pi^{-1}(B_1)}: E_1 = \pi^{-1}(B_1) \rightarrow B_1$ and $\pi|_{\pi^{-1}(B_2)}: E_2 = \pi^{-1}(B_2) \rightarrow B_2$, then the result of the theorem also holds for $\pi: E \rightarrow B$. This will imply the result in the case where B is compact, or more generally whenever B can be covered by finitely many trivializations of the vector bundle.

Let $u_1 \in H^n(E_1, E_1^\circ; R)$ and $u_2 \in H^n(E_2, E_2^\circ; R)$ be the Thom classes of the two restricted bundles. The restrictions of u_1 and u_2 to $E_1 \cap E_2 = E_\cap$ are both Thom classes for the restricted bundle over $B_1 \cap B_2$, so by uniqueness of the Thom class on E_1 and E_2 , $u_1|_{E_\cap} = u_2|_{E_\cap} = u_\cap$. Therefore, there is a well-defined cohomology class u on E whose restrictions to E_1 and E_2 are respectively u_1 and u_2 . This is clearly a Thom class because it is a Thom class fiberwise, and its uniqueness follows from the uniqueness of u_1 and u_2 , for any other cohomology class restricts to something different from u_1 on E_1 or something different from u_2 on E_2 . To show that cupping with u is an isomorphism, apply the five lemma to the diagram below, in which the rows are segments of Mayer-Vietoris sequences (and the coefficient ring R is omitted for neatness).

$$\begin{array}{ccccccccc}
H^k(E_1) \oplus H^k(E_2) & \longrightarrow & H^k(E_\cap) & \longrightarrow & H^{k+1}(E) & \longrightarrow & H^{k+1}(E_1) \oplus H^{k+1}(E_2) & \longrightarrow & H^{k+1}(E_\cap) \\
\downarrow (\smile u_1) \oplus (\smile u_2) & & \downarrow \smile u_\cap & & \downarrow \smile u & & \downarrow (\smile u_1) \oplus (\smile u_2) & & \downarrow \smile u_\cap \\
H^{k+n}(E_1, E_1^\circ) \oplus H^{k+n}(E_2, E_2^\circ) & \longrightarrow & H^\ell(E_\cap, E_\cap^\circ) & \longrightarrow & H^{\ell+1}(E) & \longrightarrow & H^{\ell+1}(E_1, E_1^\circ) \oplus H^{\ell+1}(E_2, E_2^\circ) & \longrightarrow & H^{\ell+1}(E_\cap, E_\cap^\circ)
\end{array}$$

The commutativity of this diagram follows from commutativity of the corresponding map between short exact sequences of chain complexes. Specifically, consider the diagram of chain complexes

$$\begin{array}{ccccccccc}
0 & \longrightarrow & C^*(E) & \longrightarrow & C^*(E_1) \oplus C^*(E_2) & \longrightarrow & C^*(E_\cap) & \longrightarrow & 0 \\
& & \downarrow \smile u & & \downarrow (\smile u_1) \oplus (\smile u_2) & & \downarrow \smile u_\cap & & \\
0 & \longrightarrow & C^{*+n}(E, E^\circ) & \longrightarrow & C^{*+n}(E_1, E_1^\circ) \oplus C^{*+n}(E_2, E_2^\circ) & \longrightarrow & C^{*+n}(E_\cap, E_\cap^\circ) & \longrightarrow & 0
\end{array}$$

and let $i_k: E_k \rightarrow E$ and $j_k: E_\cap \rightarrow E_k$ be the inclusions for $k = 1, 2$. Then, the diagram commutes because

$$(i_1^\# \alpha \smile u_1, i_2^\# \alpha \smile u_2) = (i_1^\#(\alpha \smile u), i_2^\#(\alpha \smile u)), \quad (j_1^\# \beta - j_2^\# \beta) \smile u_\cap = j_1^\#(\beta \smile u_1) - j_2^\#(\beta \smile u_2),$$

where $f^\#$ is the cochain map induced by f .